

Year 8 Mathematics
Mini-Test 3- Fractions, Decimals & Percentages
Resource Section (Calculators and notes allowed)



Time: 30 mins Name: _____

/30 marks

Questions 1, 2, 3 and 4 are Multiple Choice

1. Circle the option which best represents the yellow part of the rectangle? (1 mark)



- a. 0.2 **b. 40%** c. $\frac{1}{2}$ d. 0.6

2. The fraction $\frac{27}{55}$ written in exact decimal form is: (1 mark)

- a. $0.\overline{490}$ b. $0.4\dot{9}0$ **c. $0.4\overline{90}$** d. $0.49\dot{0}$

3. Circle the rational number(s). **There may be more than one.** (2 marks)

- a. $\frac{7}{8}$** b. $\sqrt{5}$ **c. 0.6** d. π

4. Circle the recurring decimal (s). **There may be more than one.** (2 marks)

- a. 0.434 b. 7.2 **c. $0.0535353\dots$** d. 0.6

Questions 5 to 10 are not multiple choice

5. Convert each of the following into fractions. (3 Marks)

a. $0.125 = \frac{1}{8}$

b. $3.2 = \frac{16}{5}$

c. $80\% = \frac{4}{5}$

6. Convert the following to decimals and use recurring decimal if necessary. (3 marks)

a. $\frac{3}{4} = 0.75$

b. 55% 0.55

c. $\frac{31}{27} = 1.\overline{148}$

7. Use appropriate notation to simplify writing these recurring decimals. (3 marks)

a. $1.33333333 =$

$1.\dot{3}$ (or $1.\overline{3}$)

b. $0.0656565 =$

$0.0\overline{65}$

c. $55.546546546 =$

$55.\overline{546}$

8. Arrange the following lists into ascending order (smallest to largest). (4 Marks)

a. $\frac{2}{5}$, 0.399, 42%, 0.382, $\frac{3}{4}$ $0.382, 0.399, \frac{2}{5}, 42\%, \frac{3}{4}$

b. $\frac{1}{5}$, 0.291, 0.302, $\frac{3}{10}$, 25% $\frac{1}{5}, 25\%, 0.291, \frac{3}{10}, 0.302$

9. State if these are rational (R) or irrational (I) (3 marks)

a. $\sqrt{2}$ (I)

b. $\sqrt{25}$ (R)

c. $\frac{1}{9}$ (R)

10. Complete the following Table (Simplify where possible) (8 marks)

Fraction	Percentage	Decimal
$\frac{1}{4}$	25%	0.25
$\frac{1}{20}$	5%	0.05
$\frac{1}{2}$	50%	0.5
$\frac{1}{10}$	10%	0.1